



Lab Sheet 2

System Response, Convolution and Correlation

Prerequisite

Before starting this section, students should know..

1. Concepts of MATLAB
2. Contents of Lab 1

Outcomes

After finishing this lab students should be able to.....

1. solve difference equation using filter function.
2. determine system response using filter function.
3. convolution and Correlation and their practical use.

Outlines of this Lab

1. Lab Session 2.1: Solution to differential Equation
2. Lab Session 2.2: Determination of system response using filter function
3. Lab Session 2.3: Convolution and Correlation of sequences
4. Lab Session 2.3: In Lab Evaluation
5. Home Work

Lab Session 2.1:

Impulse and Step Response of a system using difference equation

Consider the difference equation

$$\sum_{k=0}^N a_k y(n-k) = \sum_{m=0}^M b_m x(n-m)$$

A routine called **filter** is available in MATLAB to solve difference equation numerically , given the input and the difference equation coefficients., In its simplest form ,this routine is invoked by,

```
>> y = filter(b,a,x)
where, b=[b0, b1.....bM]
and, a= [a0, a1.....aN]
```

Example 2.1 :

Given the following difference equation

$$y(n) - y(n-1] + 0.9y(n-2) = x(n); \quad \forall n$$

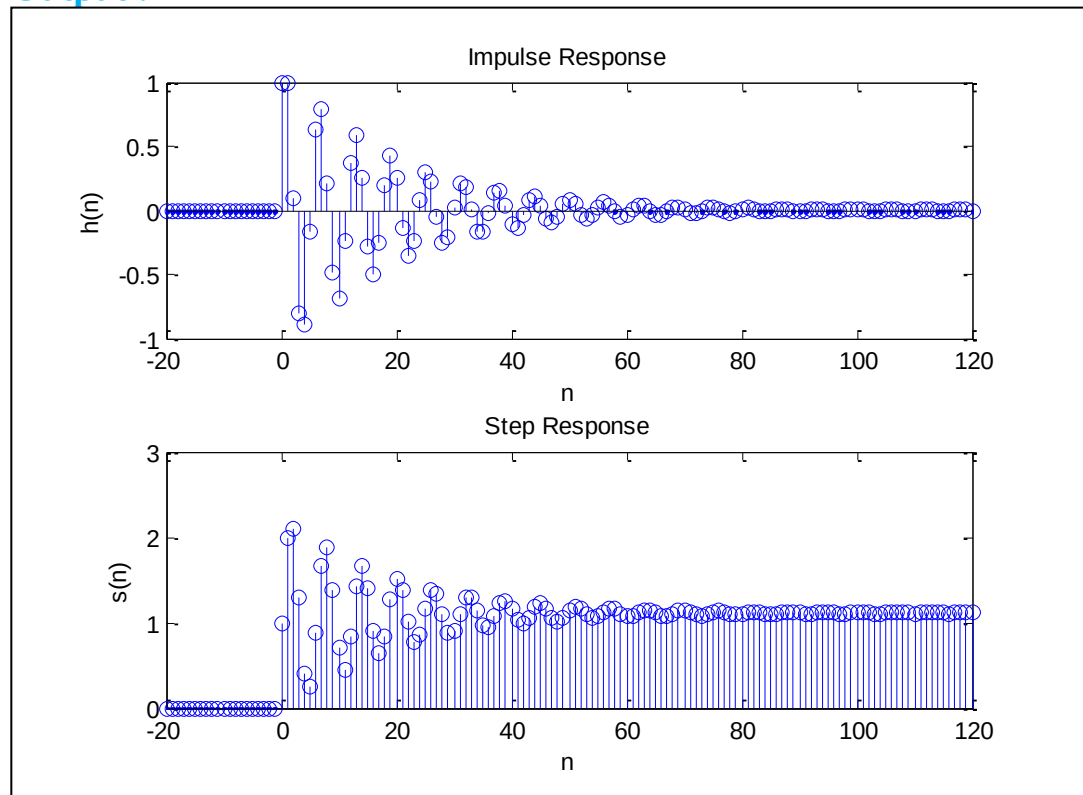
- (a) Calculate and plot the impulse response $h(n)$ at $n = [-20:100]$
- (b) Calculate and plot the step response $s(n)$ at $n = [-20:100]$
- (c) Is the system specified by $h(n)$ stable ?

Solution :

MATLAB code

```
b=[1];  
a=[1,-1,0.9];  
x=delta(0,-20,120);n=[-20:120];  
h=filter(b,a,x);  
subplot(2,1,1);stem(n,h);  
title('Impulse Response');  
xlabel('n');  
ylabel('h(n)');  
  
x1=u(0,-20,120);  
s=filter(b,a,x1);  
subplot(2,1,2);stem(n,s);  
title('Step Response');  
xlabel('n');  
ylabel('s(n)');
```

Output :



To find the stability

```
>> sum(abs(h))
```

o/p : ans = 14.8785

So , $h(n)$ is absolutely summable and so stable.

Lab Session 2.2:

System response for a given input

Example 2.2 :

Let the input and impulse response of a system are given by,

$$x(n) = u(n) - u(n-10)$$

$$\text{and, } h(n) = (0.9)^n u(n)$$

Determine the system response for $n=[-5:50]$;

Solution :

The corresponding difference equation is

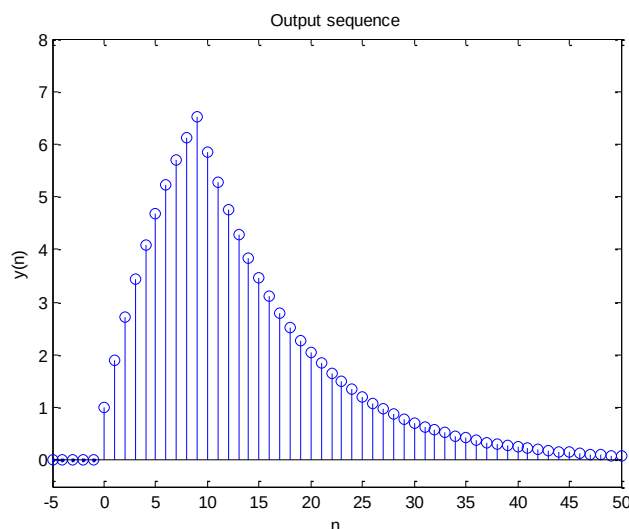
$$y(n] - 0.9y(n-1) = x(n)$$

(This is the student's part to do.)

MATLAB code

```
b=[1];  
a=[1,-0.9];  
n=[-5:50];  
x=u(0,-5,50)-u(10,-5,50);  
y=filter(b,a,x);  
stem(n,y);  
title('Output sequence');  
xlabel('n');  
ylabel('y(n)');  
axis([-5,50,-0.5,8])
```

Output :



Lab Session 2.3:

Convolution and Correlation of sequences

2.3.1 Convolution

Mathematically, a linear convolution sum is defined as

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

In general, the convolution operation is used to describe the response of an LTI system. In DSP it is an important operation and has many other uses.

MATLAB does provide a built-in function called **conv** the computes the convolution between two finite-duration sequences.

Description:

$w = \text{conv}(u,v)$ convolves vectors u and v . Algebraically, convolution is the same operation as multiplying the polynomials whose coefficients are the elements of u and v .

Thus the **conv** function assumes that the two sequences begin at $n = 0$ and is invoked by

```
>> y = conv(x,h) ;
```

Example 2.3 :

If $x = \{3 \ 11 \ 7 \ 6\}$ and $h = [4 \ -3 \ 2]$, find their convolution.

Solution :

MATLAB Code and Output:

```
>> x=[3 11 7 6];  
>> h=[4 -3 2];  
>> y=conv(x,h)
```

Output:

```
y =  
    12    35     1    25    -4    12
```

Modified function for convolution:

However, the **conv** function neither provides nor accepts any timing information if the sequences have arbitrary zero position. To solve this inconvenience, a simple extension of the **conv** function, called **conv_m**, which performs the convolution of arbitrary support sequences can now be designed as

```
function [y, ny] = conv_m(x, nx, h, nh)
% Modified convolution routine for signal processing
% -----
% [y, ny] = concolution result
% [x, nx] = first signal
% [h, nh] = second signal
nyb = nx(1) + nh(1);
nye = nx(length(x)) + nh(length(h));
ny = [nyb:nye];
y = conv(x, h);
```

Example 2.4 : If $x = \{3 \ 11 \ 7 \ 0 \ -1 \ 4 \ 2\}$, $-3 < n < 3$
and $h = \{2 \ 3 \ 0 \ -5 \ 2 \ 1\}$, $-1 < n < 4$

Find their convolution.

Solution :

MATLAB Code and Output:

```
x=[3 11 7 0 -1 4 2];nx=[-3:3];
h=[2 3 0 -5 2 1]; nh=[-1:4];
[y,ny] = conv_m(x,nx,h,nh)
subplot(3,1,1); stem(nx,x); axis([min(ny)-1 max(ny)+1 min(x)-1 max(x)+1])
subplot(3,1,2); stem(nh,h); axis([min(ny)-1 max(ny)+1 min(h)-1 max(h)+1])
subplot(3,1,3); stem(ny,y); axis([min(ny)-1 max(ny)+1 min(y)-1 max(y)+1])
```

2.3.2 Correlation of sequences

Correlation is an operation used in many applications in digital signal processing. It is a measure of the degree to which two sequences are similar. Given two real-valued sequences $x(n)$ and $y(n)$ of finite energy, the **crosscorrelation** of $x(n)$ and $y(n)$ is a sequence $r_{xy}(l)$, defined as

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y(n-l)$$

The index l is called the shift or lag parameter. The special case when $y(n) = x(n)$ is called **autocorrelation** and is defined by

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)x(n-l)$$

If we compare the convolution operation with the crosscorrelation of two sequences, we observe a close resemblance. The crosscorrelation $r_{xy}(l)$ can be put in the form

$$r_{xy}(l) = x(l) * y(-l)$$

with the autocorrelation $r_{xx}(l)$ in the form

$$r_{xx}(l) = x(l) * x(-l)$$

Therefore these correlations can be computed using the **conv** function if sequences are of finite duration.

Example 2.5 : Consider, $y(n) = x(n-2)$ where $x(n) = [3 \ 11 \ 7 \ 0 \ -1 \ 4 \ 2]$.

Find the cross-correlating $x(n)$ with $y(n)$ and discuss the result.

Solution :

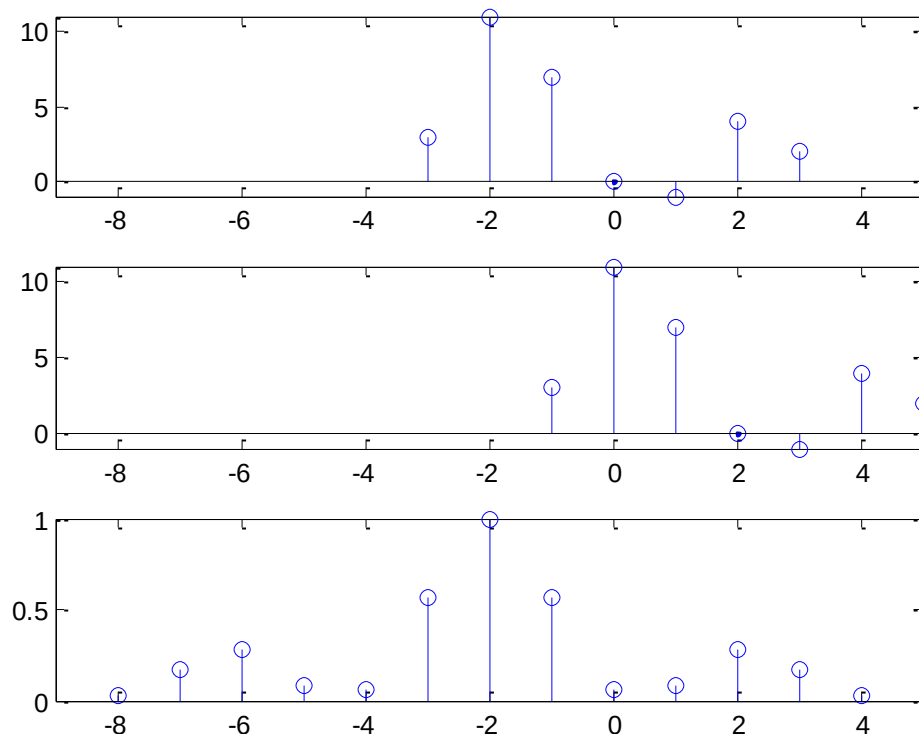
MATLAB Code:

```
x = [3,11,7,0,-1,4,2]; nx = -3:3;
[y,ny] = sigshift(x,nx,2);

[y1,ny1] = sigfold(y,ny);
[rx,rx1] = conv_m(x,nx,y1,ny1);
rx=rx/max(rx);

subplot(3,1,1);
stem(nx,x); axis([min(nx)-1 max(nx)+1 min(x) max(x)]);
subplot(3,1,2);
stem(ny,y); axis([min(ny)-1 max(ny)+1 min(y) max(y)]);
subplot(3,1,3);
stem(nxy,rx); axis([min(nxy)-1 max(nxy)+1 0 max(rx)]);
```

Output:



In the above figure, it is apparent that the CCF is very small if $l=0$ but substantial if $l=-2$. This is because $y(n)$ lags behind $x(n)$ by 2 samples and a match therefore occurs at 2 samples early.

```
figure
stem(nxy,xcorr(x,y));
```

Try the above code now and compare the new plot with the previous one.

Example 2.6 :Correlation using xcorr function:

Find and plot the correlation between the two following sequences:

$x=\{1 \ 2 \ 3 \ 4\}$
 $h=\{4 \ 3 \ 2 \ 1\};$

Use `xcorr` function.

Solution :

```
x=[1 2 3 4];  
h=[4 3 2 1];  
y=xcorr(x,h);
```

Task: Can you now rewrite the above code of the solution of Example 2.5 using **xcorr** function instead of **conv_m** so that we will get same output figure for normalized crosscorrelation?

Example 2.7:

Let $x(n) = [3, 11, 7, 0, -1, 4, 2]$ be a prototype sequence and $y(n) = x(n-2) + w(n)$ be its



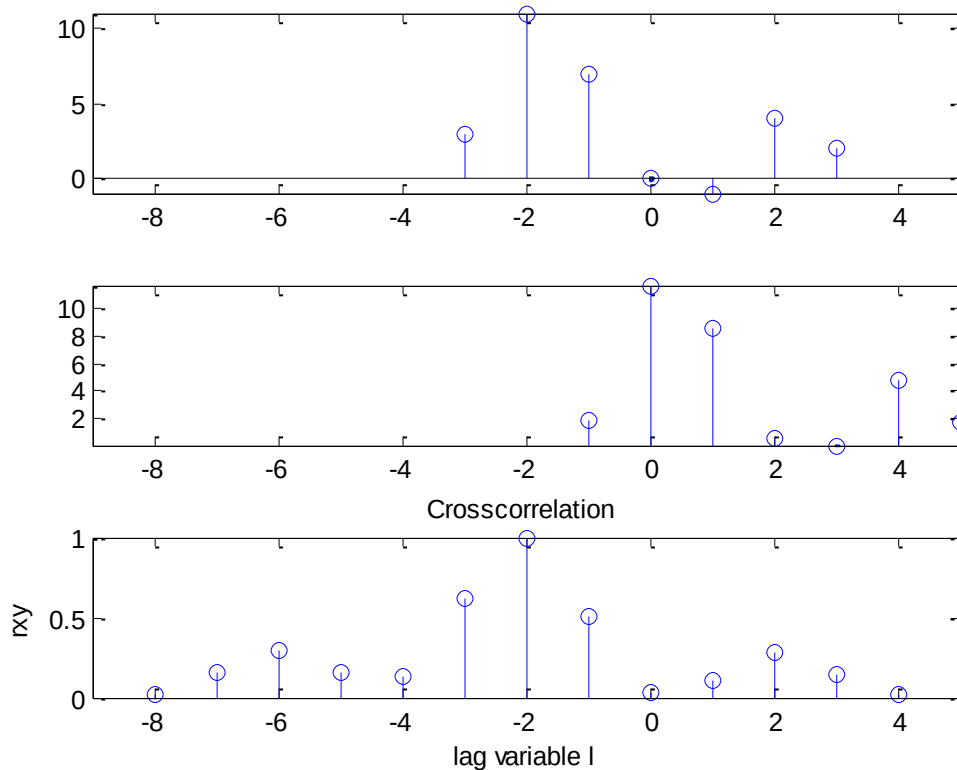
noise-corrupted and shifted version. Where $w(n)$ is Gaussian sequence with mean 0 and variance 1. Compute the crosscorrelation between $y(n)$ and $x(n)$.

Solution :

MATLAB Code:

```
x = [3, 11, 7, 0, -1, 4, 2];  
nx = [-3:3]; % given signal x(n)  
  
[y, ny] = sigshift(x, nx, 2); % obtain x(n-2)  
w = randn(1, length(y)); nw = ny; % generate w(n)  
[y, ny] = sigadd(y, ny, w, nw); % obtain y(n) = x(n-2) + w(n)  
  
rxy = xcorr(x, y); rxy = rxy / max(rxy); ny1 = -fliplr(ny);  
nxy = [(nx(1) + ny1(1)) : (nx(end) + ny1(end))];  
  
subplot(3,1,1); % plot x  
stem(nx, x); axis([min(nxy)-1 max(nxy)+1 min(x) max(x)]);  
subplot(3,1,2); % plot y  
stem(ny, y); axis([min(nxy)-1 max(nxy)+1 min(y) max(y)]);  
subplot(3,1,3); % plot rxy against lag variable  
stem(nxy, rxy); axis([min(nxy)-1 max(nxy)+1 0 max(rxy)]);  
xlabel('lag variable l'); ylabel('rxy');  
title('Crosscorrelation: noise sequence 1');
```

Output :



2.3.3 Correlation Coefficient

The correlation coefficient, denoted by r , is a measure of strength of the straight line or linear relationship between two variables. Correlation coefficient takes values between -1 and +1. A positive value implies a positive association

Correlation coefficients measure the strength of association between two variables. The **sample Pearson correlation coefficient** of two sequences $x(n)$ and $y(n)$ is given by,

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

The terms $\bar{x}$ and $\bar{y}$ are the mean values of $x(n)$ and $y(n)$ respectively.

The following points are the accepted guidelines for interpreting the correlation coefficient:

Value of coefficient	Meaning
0	No linear relationship
+1	A perfect positive linear relationship
-1	A perfect negative linear relationship

0 to 0.3	Weak positive linear relationship
0 to -0.3	Weak negative linear relationship
0.3 to 0.7	Moderate positive linear relationship
-0.3 to -0.7	Moderate negative linear relationship
0.7 to 1	Strong positive linear relationship
-0.7 to -1	Strong negative linear relationship

Example 2.8:

Write a MATLAB program to compute correlation coefficient of two sequence.

MATLAB Code:

```
X=input('Input the 1st sequence:');
Y=input('Input the 2nd sequence:');
Xm=X-mean(X);
Ym=Y-mean(Y);
rxy = sum(Xm.*Ym)/sqrt(sum(Xm.^2)*sum(Ym.^2));
disp('The correlation coefficient is:');
disp(rxy)
```

Run the above program for the following pairs of sequences and comment on the results:

- (a) $x(n) = [1 \ 2 \ 3]$ $y(n) = [1 \ 2 \ 3]$
- (b) $x(n) = [1 \ 2 \ 3]$ $y(n) = [-1 \ -2 \ -3]$
- (c) $x(n) = [1 \ 2 \ 3]$ $y(n) = [3 \ 2 \ 1]$

Lab Session 2.4:In Lab Evaluation:

IE2.1: A difference equation is given as,

$$y(n) - 0.3y(n-2) = 0.2x(n)$$

Find $y(n)$ for an input $x(n) = u(n) - u(n-3)$

IE2.2 : Two sequences are given by,

$$x(n) = [1 \ 2 \ 3] \quad y(n) = [4 \ 5 \ 6]$$

- (a) Find their crosscorrelation using `xcorr` function
- (b) Find their crosscorrelation using `conv` function
- (c) Find the correlation coefficient

Home Work :

- Q1. Invoke the following MATLAB commands and plot the result (y versus ny)
 $x_1 = [0, 0, 1, 1, 1, 1, 1]; \quad nx_1 = [-3:3];$
 $x_2 = [0, 0, 0, 1, 1, 1]; \quad nx_2 = [-3:2];$
Find the crosscorrelation of x_1 and x_2 .
- Q2. Determine the autocorrelation sequence and crosscorrelation sequence for the following sequences.

$$x(n) = (0.9)^n, \quad 0 \leq n \leq 20$$

$$y(n) = (0.8)^n, \quad -20 \leq n \leq 0$$

What is your observation ?

Q3. Consider a 20 point moving average filter

$$y(n) = \frac{1}{20}[x(n) + x(n-1) + \dots + x(n-19)]$$

- (a) Generate 200 samples of a 1 Hz sine wave sampled at 40 Hz.
- (b) Add some noise to generate a noisy signal.
- (c) Filter the noisy signal through the 20 point MA filter.
- (d) Plot each signal to display the effects of noise and smoothing.